Alex de Cock Buning

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Research Interests

Network verification, safe network automation with AI agents, software-defined networking, cloud-native and multi-cluster networking, eBPF, edge-cloud orchestration.

Education

Universidad Carlos III de Madrid — Madrid, Spain
Ph.D. in Telematics Engineering — *Mar. 2025 – Present*

Universidad Carlos III de Madrid — Madrid, Spain
M.Sc. in Internet of Things: Applied Technologies — *Completed Sep. 2024*

- Thesis: *Architecting Layer 2 Connectivities for Kubernetes Microservices: Inter-Cluster Network Slicing.* Grade: 10/10
- Awarded best master's thesis in the class (30 students)

Universidad Carlos III de Madrid — Madrid, Spain
B.Sc. in Sound and Image Engineering (Telecommunications) — *Completed May 2023*

- Thesis: *Microservices Video-on-Demand Application Deployed with Kubernetes.* Grade: 9.2/10

Research Experience

Visiting Researcher — Dec. 2025 – Present
University of Washington, advised by Prof. Ratul Mahajan — *Seattle, WA*

- Developing relational verification techniques that make network changes proposed by AI agents safe to deploy
- Built an LLM agent that proposes Kubernetes network policy changes and a relational-based verifier in OCaml that checks them
- Created a benchmark of real-world microservice and 5G scenarios to evaluate agent safety
- Leading a paper on this work for submission to ACM SIGCOMM 2027
- Contributed to research on safely optimizing eBPF programs using transformation witnesses (HotNets 2026)

Researcher (Horizon Europe-funded) — Apr. 2023 – May 2026
Universidad Carlos III de Madrid — *Madrid, Spain*

- Researcher on Horizon Europe projects in SDN, cloud-native networking, and edge-cloud orchestration, collaborating with partners across Europe
- **L2S-M**: Main developer of an open-source Kubernetes networking solution that automates a layer-2 overlay to isolate and connect pods within and across clusters.
- **HE CODECO**: Contributed to an edge-cloud orchestration framework spanning data, compute, and network, focusing on network security; built a network performance analysis tool and a virtual networks plugin

Publications

Under Submission

Alex de Cock Buning, Xiangyu Gao, Ratul Mahajan, David Walker, Francisco Valera. *Safe Agentic Network Changes via Relational Verification.* In preparation for *ACM SIGCOMM*, 2027.

Conference and Workshop Papers

Xiangyu Gao, **Alex de Cock Buning**, Xiangfeng Zhu, Jocelyn Chen, Konstantinos Kallas, Arvind Krishnamurthy, Ratul Mahajan. *Toward Safely Optimizing eBPF Programs using Transformation Witnesses.* In *Proceedings of the ACM Workshop on Hot Topics in Networks (HotNets)*, 2026.

G. Sánchez Illán, A. Muñiz Da Costa, A. Del Rio, **Alex de Cock Buning**, L. Contreras, J. Serrano, D. Jimenez, B. Nogales, I. Vidal, F. Valera, et al. *Enhanced Operation of a Media Delivery System on a Cognitive Edge-Cloud Continuum.* In *7th International Conference on Communications, Signal Processing, and their Applications (ICCSPA)*, IEEE, 2026.

Journal Articles

Alex de Cock Buning, Iván Vidal, Francisco Valera. *Architecting Multi-Cluster Layer-2 Connectivity for Cloud-Native Network Slicing. Future Internet*, 18(1):39, MDPI, 2026.

Preprints

Rute C. Sofia, Josh Salomon, Ray Carrol, ..., **Alex de Cock Buning**, et al. *Towards Scalable Federated Container Orchestration: The CODECO Approach.* arXiv:2601.13351, 2026.

Industry Experience

Cloud Advisory Intern Sep. 2021 – Sep. 2022

IBM Consulting *Madrid, Spain*

- Built a cloud-native application for evaluating cloud architecture solutions in IBM's R&D lab
- Developed a tool for analyzing mainframe applications and planning their migration to the cloud
- Took part in design-thinking sessions with clients to propose and shape new project initiatives

Skills

Languages: Go, Python, OCaml, C, Java, JavaScript
Verification : NetKAT, BDDs, equivalence checking
AI & Agents: LangGraph, LangChain, LLM evaluation
Networking & Cloud: SDN, eBPF, Kubernetes, CNI, Docker, Linux, OpenStack, OpenShift, IBM Cloud
Spoken: Spanish (native), English (C1), Dutch (A2), German (A1)